

The effects of sustained forest use on hillslope soil hydraulic conductivity in the Middle Mountains of Central Nepal

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ABSTRACT

This work investigated the multi-decadal changes in field saturated hydraulic conductivity, K_{fs} , beneath severely degraded pasture, natural forest and two mature planted *Pinus roxburghii* stands between two sets of measurements made in 1986 and 2011 at the same locations in the Middle Mountains of Central Nepal. Multiple measurements of K_{fs} were made at the four sites, both at the surface and at depths of 0.05–0.15, 0.15–0.25 and 0.25–0.50 m. The K_{fs} results were subsequently combined with rainfall intensities associated with different time intervals to infer multi-decadal changes in dominant hillslope stormflow pathways.

The widely assumed hydrological benefits of reforesting degraded land through the enhancement of near-surface permeability due to such factors as the incorporation of a greater amount of organic matter, formation of macropores, as well as root development were not observed in this study. Continued heavy use of the natural and planted forests of the Middle Mountains, particularly the removal of understory vegetation and leaf litter, and cattle grazing, are considered to be the chief causal factors of the presently observed deterioration in forest hydrological functioning. This situation is typical not only of the Middle Mountain Zone across the Himalaya but is also observed in other densely populated parts of South and South-East Asia.

The key conclusion of this work is that simply planting trees in degraded landscapes is not sufficient in itself to restore watershed hydrological functioning. Attention also needs to be given to on-going management of the reforested areas to balance product usage with watershed functions. Copyright © 2013 John Wiley & Sons, Ltd.

KEY WORDS forest degradation; hillslope hydrological response; Himalaya; reforestation; field, saturated hydraulic conductivity

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INTRODUCTION

Traditionally, the Middle Mountain Zone (or Lesser Himalaya) constitutes the most densely populated part of the Himalaya, both in Nepal (Hrabovszky and Miyan, 1987) and India (Singh *et al.*, 1984). High population pressure and heavy demands for forest products have led to widespread forest removal (Dobremez, 1976; Tucker, 1987; Singh and Singh, 1992), subsequent land degradation (Mahat *et al.*, 1986; Mahat *et al.*, 1987; Tiwari, 1988) and deterioration of streamflow regimes (Bartarya, 1989). Until comparatively recently, it was widely assumed that deforestation and overgrazing in the Himalaya were primarily responsible for the large-scale flooding and sedimentation experienced in the plains of northern India and Bangladesh (e.g. Eckholm, 1976; Nautiyal and Babor, 1985; Myers, 1986). Whilst this view is no longer tenable in the light of subsequent scientific evidence demonstrating the comparatively limited influence of land use on these *large-scale* hydrological phenomena (Bruijnzeel and Bremmer, 1989;

Ives and Messerli, 1989; Hofer, 1993; cf. Gardner and Gerrard, 2003; Hofer and Messerli, 2006), the *local* adverse hydrological effects of advanced land degradation, such as accelerated erosion, enhanced peak discharges and reduced dry-season flows (Bartarya, 1989; Bruijnzeel and Bremmer, 1989) required remedial action (cf. Negi *et al.*, 1998). At the same time, forest land had to provide a variety of goods and services (such as timber and fuel wood, plus non-timber products including water, fodder and litter for animal bedding) to local communities in support of their subsistence farming systems (Campbell and Mahat, 1975; Mahat *et al.*, 1987). A major reforestation programme was initiated in the early 1980s in the Middle Mountains of Central Nepal (Shepherd and Griffin, 1984) to address this multitude of issues. By the year 2000, ~23 000 ha of degraded pasture land and shrub land in the Districts of Sindhupalchok and Kabhre Palanchok (Central Nepal) had been converted to evergreen plantation forests (mainly *Pinus roxburghii* and *Pinus patula*) (District Forest Offices at Kabhre Palanchok and Sindhupalchok, unpublished data). The original vegetation of these areas consisted of forests of semi-evergreen broadleaf species and pines (mainly *P. roxburghii*) (Dobremez, 1976). These reforestation activities were carried out with the active participation of local communities, and most of the resulting forests were

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formally handed over to Community Forest User Groups (CFUGs) in the early 1990s. Management of these forests has been in community hands since that time, with access and usage, including animal grazing and product harvesting, carried out in accordance with approved Operational Plans. Countrywide, more than 1.65 million ha of forest, both planted and natural, has been officially handed over to CFUGs (HURDEC Nepal and Hobley, 2012). Numerous surveys have indicated that there has been a substantial increase in both forest area and forest quality across the Middle Mountains in the past two decades, thus reversing the earlier trends in deforestation in the zone (HURDEC Nepal and Hobley, 2012).

Reforestation of degraded land in the tropics and subtropics is often conducted in the expectation that disturbed streamflow regimes (commonly referred to as the 'too little–too much syndrome': Eckholm, 1976; Bartarya, 1989) will be restored by the increased rainfall absorption afforded by soil improvement after tree planting (Scott *et al.*, 2005; cf. Ilstedt *et al.*, 2007). At the same time, the water use of fast-growing tree plantations tends to be (much) higher than that of the degraded vegetation they typically replace (Scott *et al.*, 2005) – particularly where the tree roots have access to groundwater (Calder, 1992; Kallarackal and Somen, 1997). Furthermore, major improvements in the topsoil infiltration capacity of severely degraded land after tree planting may easily take several decades of *undisturbed* forest development to fully materialize (Gilmour *et al.*, 1987; Scott *et al.*, 2005; Bonell *et al.*, 2010). As such, reforesting degraded pasture or shrub land may also cause already diminished dry-season flows to become reduced even further, depending on the net balance between increases in soil water reserves afforded by improved infiltration versus decreases caused by the higher plant water uptake (Bruijnzeel, 1989; Bruijnzeel, 2004; cf. Malmer *et al.*, 2010). Naturally, such considerations assume additional importance for regional water resources management under more seasonal climatic conditions, such as those prevailing in the Middle Mountain Zone where ~80% of the annual rainfall is delivered during the main monsoon (June–September; Merz, 2004).

There is increasing evidence that soil infiltrability (Hillel, 1980) increases with time during *natural forest regrowth* after abandonment of agricultural or pasture land in the tropics (e.g. Deuchars *et al.*, 1999; Ziegler *et al.*, 2004; Zimmermann *et al.*, 2006; Hassler *et al.*, 2011). Conversely, information on the magnitude of infiltrability in tropical *tree plantations* is still scarce (see review by Ilstedt *et al.*, 2007) and usually represents a single measurement in time, rather than showing the actual development of infiltrability as the plantation matures (e.g. Mapa, 1995; Zimmermann *et al.*, 2006; Bonell *et al.*, 2010). A rare example of the latter was given by Gilmour *et al.* (1987) who reported on the contrasts of *in situ* field, saturated hydraulic conductivity (K_{fs}) (Bouwer, 1966; Talsma, 1987) in topsoils and subsoils under various land-cover types in Central Nepal. The land covers investigated ranged from heavily grazed and trampled grassland, 5-year-old

and 12-year-old pine plantations, and a relatively undisturbed natural forest. Values of K_{fs} of the deepest layer did not differ significantly between sites, suggesting comparative geological homogeneity amongst the sites. However, variability in K_{fs} increased with proximity to the surface, with maximum differences occurring in the 0–0.1 m layer, thereby causing major differences in the propensity to generate infiltration-excess overland flow (*sensu* Horton, 1933) between land-cover types. Recovery of topsoil, K_{fs} , after reforestation was evident from the increases observed for the 5-year-old and 12-year-old pine stands although values were still a long way from the infiltrability associated with the largely undisturbed natural forest. Gilmour *et al.* (1987) concluded that the observed improvements in K_{fs} relative to the degraded grassland site were due to a reduction in surface compaction by the exclusion of grazing animals (cf. Deuchars *et al.*, 1999; Hassler *et al.*, 2011), as well as improved soil aeration brought about by the enhanced activity of soil micro-flora and fauna feeding on the gradually accumulating litter layer after reforestation (cf. Lal, 1988; Zhou *et al.*, 2002). Although there are only a few studies of the adverse impacts of multi-decadal forest use on K_{fs} (e.g. Patnaik and Viridi, 1962; Bonell *et al.*, 2010), the recovery of K_{fs} following reforestation is likely to be inversely related to the intensity of forest usage. The forests (both natural and planted) of the Middle Mountain Zone along the length of the Himalayan chain are intensively used, with the lopping of branches for fuel wood, cutting of trees for timber, harvesting of understory vegetation and grasses for fodder, grazing by animals, and litter collection for animal bedding, composting and fuel all being widely practiced (Singh *et al.*, 1984; Mahat *et al.*, 1987; Gilmour and Fisher, 1991; Singh and Singh, 1992). As such, depending on the intensity of use of the replanted stands, the upward trend with time for K_{fs} envisaged by Gilmour *et al.* (1987) may continue, level off, or even decline. Even less is known about the time frame, the changes in dominant hillslope runoff generation processes and catchment-scale stormflow production are associated with the respective scenarios (cf. Bruijnzeel and Bremmer, 1989; Negi, 2002; Zimmermann and Elsenbeer, 2009; Bonell *et al.*, 2010).

In the present study, the sites of Gilmour *et al.* (1987) were revisited after more than 25 years. It was expected that re-measuring K_{fs} at the same locations and using comparable measurement techniques would provide a direct estimate of the net effect of continued forest growth and litter layer development on the one hand, and forest use and disturbance on the other, over the 25-year period between the two sets of measurements. To the best of our knowledge, this is the only study of this kind in the (sub) tropics. Next, the information on changes in K_{fs} over time was combined with regional rainfall intensity data of various temporal resolutions to infer likely changes in the dominant patterns and pathways of hillslope storm runoff response to rainfall (Chappell *et al.*, 2007) after 1986. Finally, the implications of the present findings for likely changes in regional dry-season flows as well as for forest management are discussed in the

context of land-use changes unfolding in the Middle Mountains of Nepal.

STUDY AREA

The study area (27°47'4"N, 85°42'33"E, 1660 m a.s.l.) is located in the Sindhupalchok District close to the District Headquarter town of Chautara in the Middle Mountains of Nepal, about 40 km northeast of the capital Kathmandu. The region has a complex geology that has resulted in equally complex spatial patterns of topography, climate and vegetation (Dobremez, 1976). The geology consists of phyllites, schists and quartzites in which Cambisols and Luvisols of a generally silty to clay loam have developed (Table I; cf. Maharjan, 1991; Gardner and Gerrard, 2003). The elevation varies from 800–2400 m a.s.l. Depending on the elevation, the climate is humid subtropical to warm temperate. Generally, the Middle Mountains are well watered, with annual rainfalls of 2000 mm or more, most of which (~80%) is delivered during the wet summer months ('monsoon', June–September). There is a distinct dry season that generally lasts from about October to May. However, rainfall varies with elevation and exposure to the prevailing moist air masses. The mean annual rainfall as measured at 1560 m a.s.l. in the nearby Jikhu Khola catchment between 1993 and 1998 amounted to 1487 ± 155 mm (Merz, 2004). Similar amounts were recorded in the same area at 1580 m a.s.l. between October 2010 and September 2011 by the current lead author (Ghimire *et al.*, 2012). Most rainfall events deliver less than 5 mm and at an average intensity ≤ 2.5 mm h⁻¹ (Merz *et al.*, 2006; Ghimire *et al.*, 2012). Between 1000 and 2000 m a.s.l., the natural vegetation consists of a largely evergreen mixed broadleaf forest dominated by *Schima wallichii* and various chestnuts and oaks (*Castanopsis* spp., *Quercus* spp.), patches of pine (mainly *Proxburghii*) and admixtures of *Rhododendron arboreum* above ~1500 m a.s.l. (Dobremez, 1976). Because of high population pressure and associated heavy usage of forest products by humans and animals, much of this species-rich forest had disappeared by the middle of the 20th Century (Mahat *et al.*, 1986), with most of the remaining forest either occurring on slopes that were too steep for agricultural activity or in various stages of degradation (Dobremez, 1976; cf. Figure 4). However, substantial changes in land use began occurring in the 1970s and 1980s that changed the nature of

landscape across the Middle Mountains. Prior to the 1980s, farming families in the region tended to keep relatively large herds of animals (cattle, buffalo and goats) that were open-range grazed on common land. These circumstances contributed to the deforestation and degradation of steep hillslopes that was widely reported at the time in the context of a supposed 'Himalayan environmental crisis' (Eckholm, 1976; Ives and Messerli, 1989). In the course of the 1970s, hill farmers began changing several key aspects of their farming practices. They changed their livestock management from open-range grazing of large herds to stall-feeding a small number of large animals (often one or two) and cutting and carrying fodder and animal bedding material from private and common land, a practice that still continues. A major consequence of this shift was that, particularly in some districts in Central Nepal, a large area of common land (officially Government forest land) became available for reforestation (Gilmour and Fisher, 1991). A second change commenced at about the same time when farmers began responding to a critical shortage of tree and forest products needed for their subsistence farming by planting trees. In addition, farmers commenced to allow naturally regenerated tree seedlings to remain on the periphery of their cultivated terraces and in the non-cultivated patches within the farming complex (Gilmour, 1988, 1995; Carter and Gilmour, 1989; Gilmour and Fisher, 1991; Gilmour and Nurse, 1991). The cumulative effect of these changes is that there are now substantially more trees and forests across the landscape of the Middle Mountains, on both private and common land, than there were 30 years ago (Gilmour and Fisher, 1991; HURDEC Nepal and Hobley, 2012).

The study sites were selected to represent the various stages of reforestation, ranging from heavily grazed and trampled grassland ('baseline') to a remnant broadleaved forest whose tree layer was heavily disturbed ('semi-natural forest'). The oldest surviving reforestation stands in the area at the time the measurements were taken (February 2011) had an age of 36 years. The visual inspection of the greenness of the leaves and of the presence/absence of external diseases suggested that the planted forests were healthy enough at the study sites. Previously, the area had been subject to substantial deforestation dating back more than a century (Mahat *et al.*, 1986).

Of the five original sites of Gilmour *et al.* (1987), four were selected for re-measurement of K_{fs} . The soil surface

Table I. Selected physical and chemical properties of the soil at 0–0.15 m depth associated with the different land-cover types under study near Chautara, Central Nepal. Values listed are the means ($\pm$ SD) of six samples per site. Values sharing at least one same superscript are not significantly different from each other and vice versa.

Variable	Degraded pasture [Site-1]	Pine forest [Site-3]	Pine forest [Site-4]	Semi-natural forest [Site-5]
Bulk density [g cm ⁻³]	1.30 ^a ($\pm$ 0.05)	1.20 ^a ($\pm$ 0.09)	0.97 ^b ($\pm$ 0.06)	0.90 ^b ($\pm$ 0.18)
Clay [%]	34 ^c ($\pm$ 7)	30 ^c ($\pm$ 6)	39 ^c ($\pm$ 11)	20 ^c ($\pm$ 10)
Sand [%]	21 ^{f,g} ($\pm$ 6)	30 ^f ($\pm$ 5)	12 ^g ($\pm$ 7)	25 ^f ($\pm$ 12)
Silt [%]	45 ^h ($\pm$ 11)	40 ^h ($\pm$ 8)	49 ^h ($\pm$ 11)	55 ^h ($\pm$ 10)
SOC [%]	3.6 ⁱ ($\pm$ 0.77)	4.3 ⁱ ($\pm$ 0.98)	5.3 ^{i,j} ($\pm$ 1.2)	6.2 ^j ($\pm$ 1.1)

SOC, soil organic carbon.

of the second site (i.e. Site 2 in Table I of Gilmour *et al.*, 1987) had numerous large stones, thereby rendering it difficult to take consistent measurements of K_{fs} . The four investigated sites (with site numbers in brackets corresponding with those of Gilmour *et al.*, 1987) can be characterized as follows:

- Degraded pasture land (Site-1): Originally deforested more than 125 years ago and already heavily grazed and trampled with widespread soil erosion in 1986. The current site condition differs only slightly from what it was in 1986 (Figure 1). Numerous foot paths are now distributed across the slope. The site is also more severely over-grazed and patches of soil surface are exposed. Little or no vegetation exists. The results of the re-measurements are considered to represent the net effect of 25 years of additional cattle grazing and trampling pressure by both cattle and humans.
- Pine forest (Site-3): Planted with *P. roxburghii* in 1974 and aged 36 years in 2011. Site-3 was similar to Site-1 prior to reforestation but grazing had been excluded for 12 years by 1986, although litter was regularly collected. The surface condition in 2011 was quite different from that in 1986 (Figure 2). Since 1990, there has been an increased cattle grazing. In addition, most of the understory has been removed and grass is cropped to ground level. The results of the re-measurements are considered to represent the net effect of more than 20 years of continuous heavy cattle grazing and trampling.
- Pine forest (Site-4): Planted with *P. roxburghii* in 1974 and aged 36 years in 2011. Location is adjacent to Site-1. The vegetation prior to planting consisted of low shrubs and herbs about 40 cm high plus a few scattered remnants of the original forest in 1986 (Figure 3). In 1990, the local people started to collect the litter layer for animal bedding. Nearly 50% of the planted trees were removed for timber and fuel wood between 1990 and 2010 in accordance with an approved community forest Operational Plan. However, grazing animals are generally excluded with the help of a forest watcher. The soil surface is well covered with broadleaved shrubs. The results of the re-measurements are considered to reflect the net effect of more than 20 years of litter collection and harvesting understory vegetation.
- Near-natural forest (Site-5): Back in 1986, this forest was considered to be in a near-natural state as harvesting was limited to some leafy materials for animal bedding and fodder. The current condition of this site is very different from what it was in 1986 (Figure 4). Extensive tree loss has occurred, in the meantime, although the soil surface is still well covered with a dense broadleaved understory as well as a litter layer. Grazing animals and litter collection are excluded with the help of a forest watcher. The results of the re-measurements are considered to reflect the net effect of the gradual transformation of a nearly undisturbed natural forest to a shrub land with scattered remnant trees, particularly the effect of manual logging and ground-based removal of timber.

MATERIALS AND METHODS

Sampling design

On the basis of interviews with local people during the field campaign in February 2011, the sites can be considered to have been free from natural disturbances such as landslips. Hence, any improvement in K_{fs} within the reforested sites over 25 years that have elapsed since the first round of measurements might be attributed largely to the effect of reforestation (i.e. the building up of the litter layer and incorporation of soil organic matter; cf. Lavelle and Spain, 2001), and any decrease in K_{fs} to site disturbance by humans and animals (e.g. trampling pressure, grazing, and litter collection) and thus to increased soil surface exposure to erosive crown drip (cf. Wiersum, 1985; Brandt, 1988; Hall and Calder, 1993).

Multiple measurements of field K_{fs} were made at the four Chautara sites, both at the surface and at depths of 0.05–0.15, 0.15–0.25 and 0.25–0.5 m. It was envisaged that the measurements made at the latter depth intervals would provide sufficient information to characterize the nature of water movement through the soil profiles on the prevailing steep slopes. For all depths except at the surface, a stratified sampling approach was used, whereas measurements were made at the plot scale. First, two 45 m long lines were laid out within each plot along and perpendicular to the hillslope gradient. This was carried out to avoid the clustering of sampling in one direction only and to distribute the samples both along and across the hillslope. Further, K_{fs} was measured at 3 m intervals, which resulted in 16 measurements in each direction. For the measurement of surface K_{fs} , 5–6 replications were taken per site via random sampling using a disc permeameter. The number of K_{fs} measurements made at the surface was much less than for the other depths because of the practical limitations of using a disc permeameter on steeply sloping land (Smettem K.R.J., personal communication).

K_{fs} measurements

A disc permeameter (Perroux and White, 1988; McKenzie *et al.*, 2002) was used for the measurement of surface K_{fs} . Before conducting the measurements, any straw and stubble present were removed from the soil surface with the least amount of surface disturbance. A steel ring of 25.5 cm diameter and 3 cm height was then inserted into the soil for about 4 mm. In order to improve the contact between the disc and the soil surface, a 5 mm thick layer of fine sand (size <2 mm) was placed on the soil surface inside the ring. After preparing the soil surface in this way, the apparatus was placed on the ring. The rates of water discharge through the disc, as inferred from changes in the water levels in the storage tower of the apparatus, were recorded until steady-state flow was reached. K_{fs} was then calculated along the lines of McKenzie *et al.* (2002):

$$K_{fs} = I - \frac{4bS_o^2}{\pi r(\theta_n - \theta_o)} \quad (1)$$

where θ_n is the initial volumetric soil moisture content of the *in situ* soil before the experiment; θ_o is the *in situ*

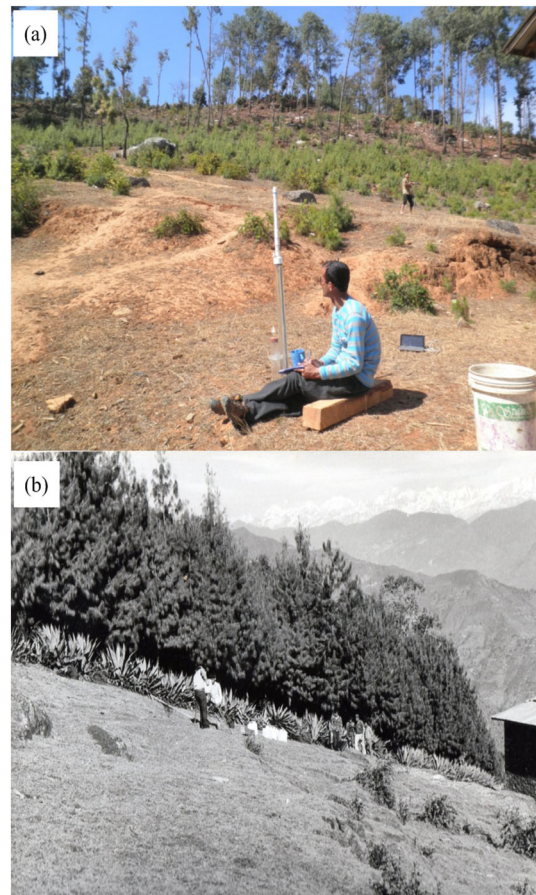


Figure 1. Degraded pasture (Site-1) near Chautara, Central Nepal in: (a) 2011 (Colour), and (b) 1986 (B/W).

volumetric soil moisture content after the experiment; b is a constant (0.55); S_o , the sorptivity obtained by plotting cumulative infiltration volume versus the square root of time since the start of infiltration; r , the radius of the disc permeameter base (12.75 cm); and I , the steady-state infiltration rate, calculated as

$$I = \frac{q}{\pi r^2} \quad (2)$$

where q is the slope of the plot of cumulative infiltration versus time, after the infiltration rate has reached a steady state, with r as defined previously.

For the measurement of K_{fs} in deeper soil layers, and in line with the previous work of Gilmour *et al.* (1987), the constant-head well permeameter (CHWP) was used (Talsma and Hallam, 1980). The use of the CHWP was restricted to the dry season to minimize errors from smearing of the auger hole walls (Chappell and Lancaster, 2007). The experimental procedure involved augering a cylindrical hole ($a=4$ cm) to the desired depth. After augering the hole, any sealing of pores in the column walls because of the augering was negated by brushing the walls with a small metal brush. The auger hole was pre-wetted for 20 min before taking the measurement to achieve perimeter saturation as described by Talsma and Hallam (1980). The CHWP was then inserted to the required depth and the flow measured until a steady state was reached. K_{fs} values were calculated from the measurements using

Equation 11 of Reynolds *et al.* (1983):

$$K_{fs} = \frac{CQ_t}{2\pi H^2 \left[1 + \frac{C}{2} \left(\frac{a}{H} \right)^2 \right]} \quad (3)$$

where Q_t is the total flow out of the cylindrical hole; H , the constant height of ponded water in the well; a , the radius of the well; and C , the dimensionless shape factor calculated as:

$$C = \sinh^{-1} \left(\frac{H}{a} \right) - \sqrt{\left(\left(\frac{a}{H} \right)^2 + 1 \right)} + \frac{a}{H} \quad (4)$$

Inferring dominant hillslope hydrological pathways during rainfall

A growing body of hillslope hydrological and small catchment studies in the tropics and subtropics suggests that the most important parameters governing the way water runs off during and shortly after rainfall events (the so-called dominant stormflow pathways; Bonell, 2005) include the vertical distribution of K_{fs} and prevailing rainfall intensities (Bonell *et al.*, 1983; Gilmour *et al.*, 1987; Wenzel *et al.*, 1998; Elsenbeer *et al.*, 1999; Godsey and Elsenbeer, 2002; Ziegler *et al.*, 2006; Zimmermann *et al.*, 2006; Zimmermann and Elsenbeer, 2009; Bonell *et al.*, 2010; Germer *et al.*, 2010; Hassler *et al.*, 2011).



Figure 2. Pine reforestation site (Site-3) near Chautara, Central Nepal in: (a) 2011 (Colour), and (b) 1986 (B/W).

In order to infer the dominant stormflow pathway at each site, selected percentiles of maximum rainfall intensities $I_{\max}$ (e.g. over 5 or 30 min: $I_{5\max}$ and $I_{30\max}$), as measured at Dhulikhel (~22 km towards the south-east of the study sites, see succeeding text), were superimposed on the respective box plots of K_{fs} . Although short and intense storms can lead to the occurrence of infiltration-excess overland flow (IOF; Horton, 1933), longer duration rainfall of low intensity can produce subsurface stormflow (SSF) and saturation-excess overland flow (SOF) (Bonell, 2005). The latter is usually generated in low-lying areas at footslopes, around streams and in depressions, but SOF may also become widespread on hillslopes where rainfall intensities exceed the transmission capacity of a soil layer at shallow depths (Bonell *et al.*, 1983).

Rainfall measurements

The rainfall intensity data used in the characterization of the dominant hillslope hydrological pathways in the Chautara plots and changes therein as a function of ecosystem degradation were recorded at Dhulikhel during the monsoon periods of 2010 and 2011. Although made at a location some 22 km towards the south-east, these data were considered to be typical of the rainfall characteristics in the Chautara area as they were collected at a similar elevation (1580 m a.s.l. vs 1660 m a.s.l.) and under the same environmental conditions. At any rate, they are likely to be more representative than the rainfall records for

Kathmandu (being the only data available at the time) that were used in the previous analysis by Gilmour *et al.* (1987).

Rainfall was recorded using a tipping-bucket rain gauge (Rain Collector II, Davis Instruments, USA; 0.2 mm per tip) at 5-min intervals. Rainfall events were defined as events totaling at least 5 mm (Negishi *et al.*, 2006) and separated from each other by a dry period of at least 3 h. The maximum 5-min rainfall amounts ($I_{5\max}$, expressed as equivalent hourly rainfall intensity) were determined by calculating the maximum precipitation over the corresponding time interval for each event. In addition, the maximum 10 ($I_{10\max}$), 15 ($I_{15\max}$), 30 ($I_{30\max}$) and 60 min ($I_{60\max}$) rainfall intensities (all expressed as equivalent hourly rainfall intensities) were derived as well.

Additional soil physical and chemical properties

Additional basic soil physical and chemical properties such as dry bulk density, texture and soil organic carbon content (SOC) were determined in order to explore their possible influence on the magnitude of near-surface K_{fs} next to the influence of land/forest management.

Soil bulk density was measured in each plot at six randomly selected points at a depth of 0–0.15 m on undisturbed soil cores (294 cm³; Blake and Hartge, 1986). At the same depth and sampling points, soil textural composition was determined using the hydrometer method (Gee and Bauder, 1986). SOC was determined using the dry combustion method (Nelson and Sommers, 1982).



Figure 3. Pine reforestation site (Site-4) near Chautara, Central Nepal in: (a) 2011 (Colour), and (b) 1986 (B/W).

*Comparison with the results obtained by Gilmour *et al.* (1987)*

In order to quantify the net changes in field, saturated hydraulic conductivity caused by the multi-decadal use and disturbance of the Chautara forests, the presently obtained log-mean K_{fs} values and inferred dominant hillslope hydrological flow paths were compared with those for 1986 of Gilmour *et al.* (1987) for the selected soil layers. As the depth ranges used in the current field experiment were not entirely identical with those adopted by Gilmour *et al.* (1987), only those K_{fs} values representing soil layers at similar depth intervals were selected for comparison. K_{fs} measurements for the following depth intervals were used: (1) Surface; (2) 0.05–0.15 m; and (3) 0.25–0.50 m for comparison with Gilmour's (1) Surface; (2) 0.10–0.20 m; and (3) 0.20–0.50 m, respectively. For the deeper soil layers (i.e. >0.2 m depth), as previously indicated, Gilmour *et al.* (1987) used the same instrument as the one used in the present study (i.e. the Talsma CHWP). However, the instrument used for the measurement of surface K_{fs} in the present study (i.e. the disc permeameter) differed from the ring infiltrometer

(Talsma, 1987), as used by Gilmour *et al.* (1987). The implications of this will be discussed later.

Statistical analysis

Because of the non-normality of the K_{fs} data, non-parametric tests were applied for statistical analysis. Differences in K_{fs} between 1987 and 2011 for the selected layers were statistically examined using the Mann–Whitney *U*-test. Differences were taken to be significant when $p < 0.05$. The surface K_{fs} of all sites are excluded from statistical analysis because of the small size ($n = 5–6$ in 2011). For the statistical comparison of soil characteristics (bulk density, texture and SOC) between the sites, Kruskal–Wallis test (Kruskal and Wallis, 1952) was used. If the Kruskal–Wallis test indicated a significant difference, the Mann–Whitney *U*-test with Bonferroni correction was applied to account for the multiple comparisons across sites.

RESULTS

Field, saturated hydraulic conductivity, K_{fs}

The descriptive statistics for field, saturated hydraulic conductivity, K_{fs} for the various depth intervals at each site are presented in Table II and are plotted in Figure 5.

As expected, the median surface K_{fs} was lowest for the degraded pasture (Site-1) and highest for the degraded natural forest (Site-5), the median values being 32 and 228 mm h⁻¹, respectively, and so these differ by an order of magnitude (Table II). K_{fs} decreased with depth throughout the profile underneath the natural forest but values in the two pine stands below a depth of 0.05 m did not vary much. K_{fs} values in the reforestation stands were slightly higher at 0.15–0.25 m depth compared with 0.05–0.15 m and attained their lowest values at a depth of 0.25–0.50 m, similar to the situation observed in the natural forest (Figure 5). Conversely, K_{fs} in the degraded pasture first increased to a depth of 0.25 m and then started to decrease. For the 0.05–0.15 m depth interval, the most noticeable difference between sites is the median K_{fs} value at Site-3 (mature pine forest), which is markedly lower than in any of the other sites (Table II). Furthermore, K_{fs} values in the 0.05–0.15 and 0.15–0.25 m layers differed considerably between sites but less between the two depths at a given site, with no marked difference between the two depths at Site-3, a slight increase in K_{fs} with depth at Site-4, and a slight decrease with depth at Site-5 (Table II and Figure 5). At 0.50 m depth, however, the differences in K_{fs} between the respective land-cover types became smaller, indicating the limited influence of trampling (cattle grazing and human pressure) on the deeper soil layers.

A comparison of the present log-mean K_{fs} values with those of Gilmour *et al.* (1987) is presented in Table III. The data clearly show that only a small reduction in surface K_{fs} was detected at the degraded grassland site (Site-1) after the additional 25 years of human trampling and cattle grazing, in contrast to the relatively larger reduction

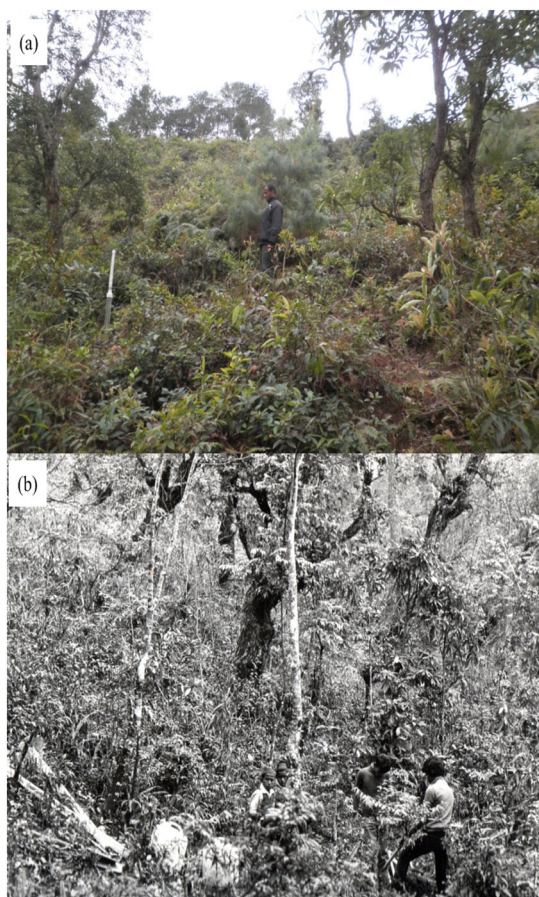


Figure 4. Near-natural forest site (Site-5) near Chautara, Central Nepal in: (a) 2011 (Colour), and 1986 (B/W).

($\sim 40 \text{ mm h}^{-1}$) observed at a depth of 0.05–0.15 m. On the other hand, dramatic reductions in K_{fs} under both planted and natural forest were observed following increased forest use and disturbance since 1986. For example, reductions in K_{fs} of as much as $\sim 200 \text{ mm h}^{-1}$ were derived both for the surface and the shallow subsurface soil ($< 0.25 \text{ m}$) at Site-5. Similarly, an even more pronounced reduction at Site-4 (pine forest) of $\sim 430 \text{ mm h}^{-1}$ at the surface was observed *versus* $\sim 130 \text{ mm h}^{-1}$ at shallower depth ($< 0.25 \text{ m}$). Reductions at Site-3 (which had lower initial K_{fs} values to start with in 1986, presumably because of its more intensive land-use history prior to planting compared with Site-4; Gilmour *et al.*, 1987) were less pronounced than those for Site-3, viz. $\sim 90 \text{ mm h}^{-1}$, both at the surface and at shallow depth ($< 0.25 \text{ m}$). Values of K_{fs} at 0.25–0.50 m depth were lower in 2011 at all four sites, although the respective reductions are much smaller than those observed at depths $< 0.25 \text{ m}$ (Table III). Furthermore, K_{fs} for the respective soil layers, 0.05–0.15 and 0.25–0.5 m depths at the investigated sites, differed significantly ($p < 0.05$) between the 1987 and 2011 data sets (Table III).

Rainfall intensities and inferred dominant hillslope runoff pathways

A total of 99 storm events were recorded at Dhulikhel during the summer monsoons of 2010 and 2011. The maximum equivalent hourly intensities for the observed 5-min,

10-min, 15-min, 30-min, and 60-min rainfall amounts ($I_{5\text{max}} - I_{60\text{max}}$) for all storms were 88.8, 82.8, 62.4, 47.2, and 39.6 mm h^{-1} , respectively. The corresponding median values were 26.4, 19.2, 15.6, 10.6, and 7.6 mm h^{-1} . For each rainfall intensity class, intensities were low for the majority of events (Figure 6). However, the higher-intensity storms contributed substantially to the bulk of overall monsoon precipitation totals (Figure 6). The median, 75% percentile, 95% percentile, and maximum values of $I_{5\text{max}}$ and $I_{30\text{max}}$ were selected as indices for inferring the dominant hillslope runoff pathways during rainfall following Zimmermann and Elsenbeer (2009), and these have been plotted in conjunction with the results obtained for K_{fs} for the respective land covers in Figure 5.

It is evident from Figure 5(a) that for the degraded pasture site, the upper quartile of $I_{5\text{max}}$ exceeds the entire range of measured surface K_{fs} values and that IOF is the inferred dominant stormflow pathway. Furthermore, 5-min rainfall totals between the 95th percentile and the maximum value continue to exceed the median K_{fs} for this site at all depths. Thus, vertical percolation will be limited for all rainfall events with these intensities (Figure 5(a)). However, a noticeable shift from IOF-dominated runoff to unimpeded infiltration occurs when considering the maximum rainfall intensity over 30 min, i.e. moving from $I_{5\text{max}}$ to $I_{30\text{max}}$ (Figure 5(b)). The presently inferred dominant stormflow pathway for the degraded pasture is similar to that suggested by the K_{fs} results obtained in 1986 for this site by Gilmour *et al.* (1987) (Figure 7(a,b)). This suggests that there has been only limited additional soil degradation at this site over the last 25 years.

Despite the lower topsoil K_{fs} values determined for the two planted forest sites compared with those of the degraded natural forest, the median surface K_{fs} values beneath the pines were still above (or nearly equal to) the most intense rainfalls (Figure 5(c,e)), indicating that IOF in the plantations must still be considered a rare phenomenon. However, subsurface median K_{fs} values at Site-3 between 0.05 and 0.50 m are so much lower now than in 1986 (Table III), that they are well below the upper quartile of $I_{5\text{max}}$ (Figure 5c) and the 95th percentile of $I_{30\text{max}}$ (Figure 5(d)). Hence, perched water tables at depths as shallow as 0.05–0.25 m are likely to be generated more frequently during rain events at Site-3. Depending on the depth of occurrence of such temporary groundwater tables, the water may run off in the form of shallow SSF or even as hillside SOF. Similar patterns were derived for Site-4, although in view of the generally higher K_{fs} at all depths compared with Site-3 due to its more benign land-use history (Table II), the development of SSF at Site-4 is expected to be less frequent and SOF all but non-existent (Figure 5(e)).

Overall, when comparing the present situation with that in 1986 for the two reforestation stands (Figure 7(c–f)), a distinct change in hillside runoff pathways at these two sites is indicated. Such circumstances reflect the increased human trampling and cattle grazing pressure during the last 20–25 years. The 0.05–0.15 m soil layer at Site-3 is now more likely to develop a perched water table and thus SSF

Table II. Descriptive statistics for field, saturated hydraulic conductivities K_{fs} [mm h^{-1}] associated with contrasting land-cover types at different depths near Chautara, Central Nepal.

	Surface	(0.05–0.15 m)	(0.15–0.25 m)	(0.25–0.50 m)
Degraded pasture [Site-1]				
Mean	33	48	59	45
Median	32	43	50	43
Standard deviation	5	21	30	22
Minimum	28	14	21	13
Maximum	40	96	146	100
Sample size (n)	5	32	32	32
Pine forest [Site-3]				
Mean	114	31	34	27
Median	83	25	33	21
Standard deviation	73	15	13	37
Minimum	32	11	18	9
Maximum	225	86	75	225
Sample size (n)	6	32	32	32
Pine forest [Site-4]				
Mean	117	71	81	62
Median	94	64	77	55
Standard deviation	90	26	34	33
Minimum	33	29	32	6
Maximum	279	125	193	157
Sample size (n)	6	32	32	32
Semi-natural forest [Site-5]				
Mean	204	147	128	68
Median	228	137	126	66
Standard deviation	109	104	53	24
Minimum	50	29	29	36
Maximum	340	570	214	133
Sample size (n)	6	32	32	32

(occasionally supplemented by SOF) during most maximum 5-min rainfall intensities, whereas the same soil layer was transmissive to all maximum 5-min rainfall intensities 25 years ago (Figure 7(c)). Similarly, perched water tables and thus SSF (but not SOF) are likely to develop more frequently in the 0.05–0.15 and 0.25–0.50 m soil layers at Site-4 for which the historical data suggest they were (much) more permeable 25 years ago and thus capable of accommodating all 5-min rainfalls (Figure 7(e)).

By contrast, despite being gradually degraded in terms of tree biomass (including the manual removal of all larger trees) over the last 25 years (Figure 4), the (semi-)natural forest of Site-5 still showed considerably higher K_{fs} values down to 0.25–0.50 m. Five-minute maximum rainfall intensities did not exceed the median K_{fs} down to 0.15–0.25 m depth (Figure 5). Consequently, the natural forest is still characterized by predominantly vertical percolation down to 0.15–0.25 m for all rainfall intensities. However, a perched water table is likely to develop at 0.25–0.50 m depth during the most intense rainfalls (Figure 5). A comparison of inferred dominant hydrological flow paths based on the new K_{fs} data with those of Gilmour *et al.* (1987) suggests no significant change in the dominant storm-flow pathways as yet in the (semi-)natural forest (Figure 7(g, h)), despite the observed major reductions in surface and subsurface K_{fs} (Table III).

DISCUSSION

Uncertainty in K_{fs}

Several studies have demonstrated that the use of different techniques for the determination of K_{fs} (for example, soil cores versus well permeametry) may produce significantly different results (e.g. Talsma and Hallam, 1980; Davis *et al.*, 1999; Chappell and Lancaster, 2007). Smearing of the auger hole and thus pore closure and/or disruption of pore continuity as a result of augering are reasons commonly given for the underestimation of hydraulic conductivity using well permeametry compared with core-based values (Talsma, 1987; Davis *et al.*, 1999; Chappell and Lancaster, 2007). Because the K_{fs} data for the Chautara sites were collected during the dry season and the internal walls of the augered holes were brushed carefully, it was assumed that the smearing and blocking of soil pores and thus the effect on K_{fs} was minimized.

Values for the subsoil K_{fs} were determined using similar instruments and sample sizes in the two studies. Therefore, it is expected that the presently obtained results for subsoil K_{fs} are comparable with those of Gilmour *et al.* (1987). However, for the surface K_{fs} , the instrument used in 2011 (disc permeameter) differed from that used in 1986 (ring infiltrometer; Dunin, 1976; Talsma, 1987). Despite all care being taken during the 1986 survey, one cannot exclude a small risk of surface soil disturbance during ring insertion that may have had an effect on the results. Nevertheless,

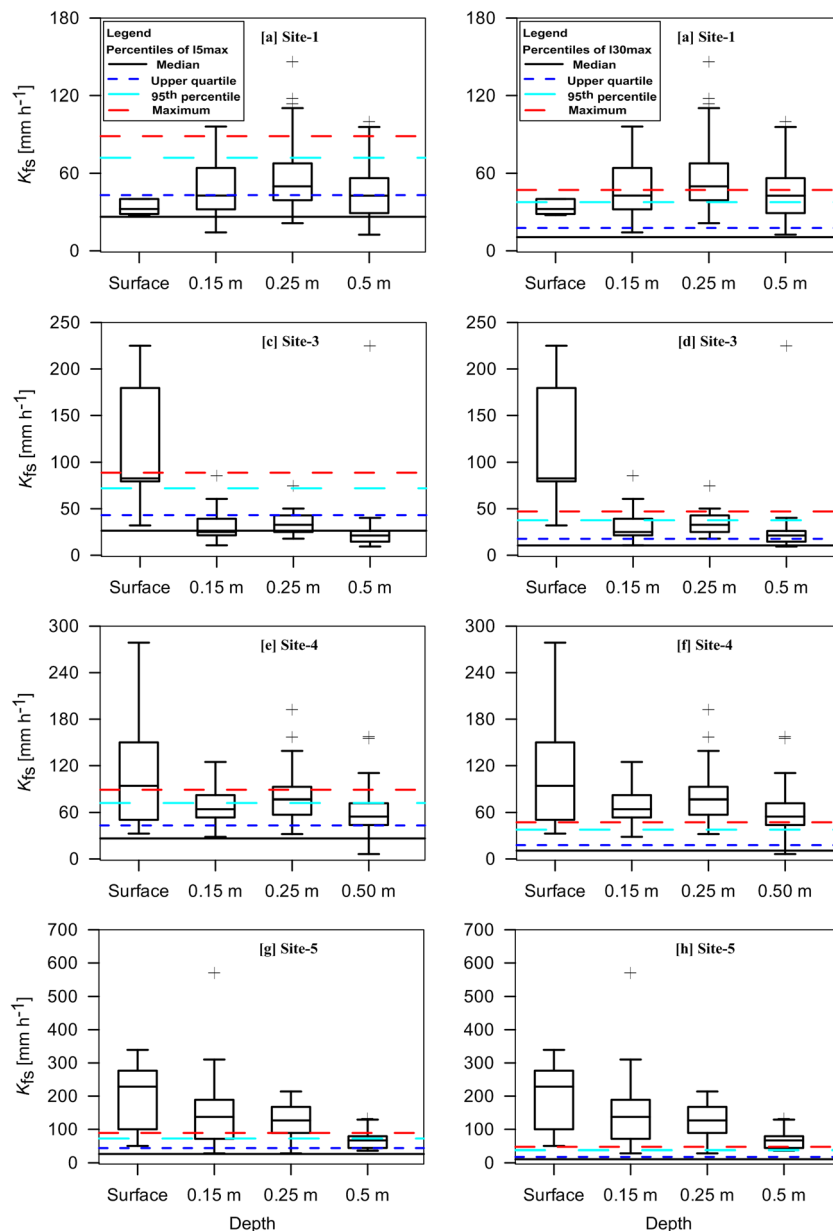


Figure 5. Box plots for: (a,b) Degraded pasture (Site-1), (c,d) Pine forest (Site-3), (e,f) Pine forest (Site-4), and (g,h) natural forest (Site-5). The solid horizontal line represents the median rainfall intensity and dashed lines the 75% percentile, 95% percentile and maximum values of $I_{5\max}$ (on the left: a,c, e,g) and $I_{30\max}$ (on the right: b,d,f,h) rainfall intensities, respectively.

in view of the very similar measurement scales of the two instruments (ring diameters of 0.30 and 0.26 m, respectively), the respective measurements should be largely comparable. Some support for this contention comes from the fact that K_{fs} trends found in the 2011 survey for the surface and for the 0.05–0.15 m depth interval are similar to those of Gilmour *et al.* (1987) at all sites. Hence the K_{fs} measured at the surface may serve as an indicator of the change in condition despite the smaller sample size of the 2011 measurements.

Effects of multi-decadal land use on K_{fs}

Degraded pasture. Decreases in K_{fs} at shallow depths associated with the conversion of tropical forest to pasture are well documented (e.g. Alegre and Cassel, 1996; Tomasella and Hodnett, 1996; Deuchars *et al.*, 1999;

Zimmermann *et al.*, 2006; Tobón *et al.*, 2010). Most of the often substantial reductions in K_{fs} appear to have occurred during the first 3–5 years after forest clearing (Alegre and Cassel, 1996; Martinez and Zinck, 2004), but the impact of continued multi-decadal compaction by grazing animals is less well known. Gilmour *et al.* (1987) considered their degraded pasture site to have been in use for more than a century prior to the K_{fs} measurements in 1986. As such, by the time of the present study (2011), the site had experienced ~125 years of intensive grazing (cf. Figure 1). In view of the similarity in the log-mean values of surface K_{fs} determined in 1986 (39 mm h^{-1}) and 2011 (33 mm h^{-1}), it would seem as though the continued intensive use of the land over the last 25 years had limited additional impact on surface K_{fs} . The presently found values for surface K_{fs} are comparable with the 41 mm h^{-1} reported by Alegre and Cassel (1996) for an intensively grazed 5-year-old

Table III. Comparisons of the presently derived log-mean field, saturated hydraulic conductivities K_{fs} [mm h⁻¹] for contrasting land-cover types near Chautara, Central Nepal, with the corresponding values of Gilmour *et al.* (1987).

	Log-mean Hydraulic Conductivity K_{fs} [mm h ⁻¹]		Difference
	Present study	Gilmour <i>et al.</i> (1987)	
Surface (0–0.1 m)			
Degraded pasture [Site-1]	33	39	—
Pine forest [Site-3]	96	183	—
Pine forest [Site-4]	93	524	—
Natural forest [Site-5]	173	370	—
0.05–0.15 m (0.10–0.20 m)			
Degraded pasture [Site-1]	44	87	s.
Pine forest [Site-3]	29	117	s.
Pine forest [Site-4]	67	195	s.
Natural forest [Site-5]	130	310	s.
0.25–0.50 m (0.20–0.50 m)			
Degraded pasture [Site-1]	40	63	s.
Pine forest [Site-3]	20	38	s.
Pine forest [Site-4]	54	100	s.
Semi-natural forest [Site-5]	65	96	s.

Values in parentheses are the measurement depths adopted by Gilmour *et al.* (1987).
s. symbolises “significant”.

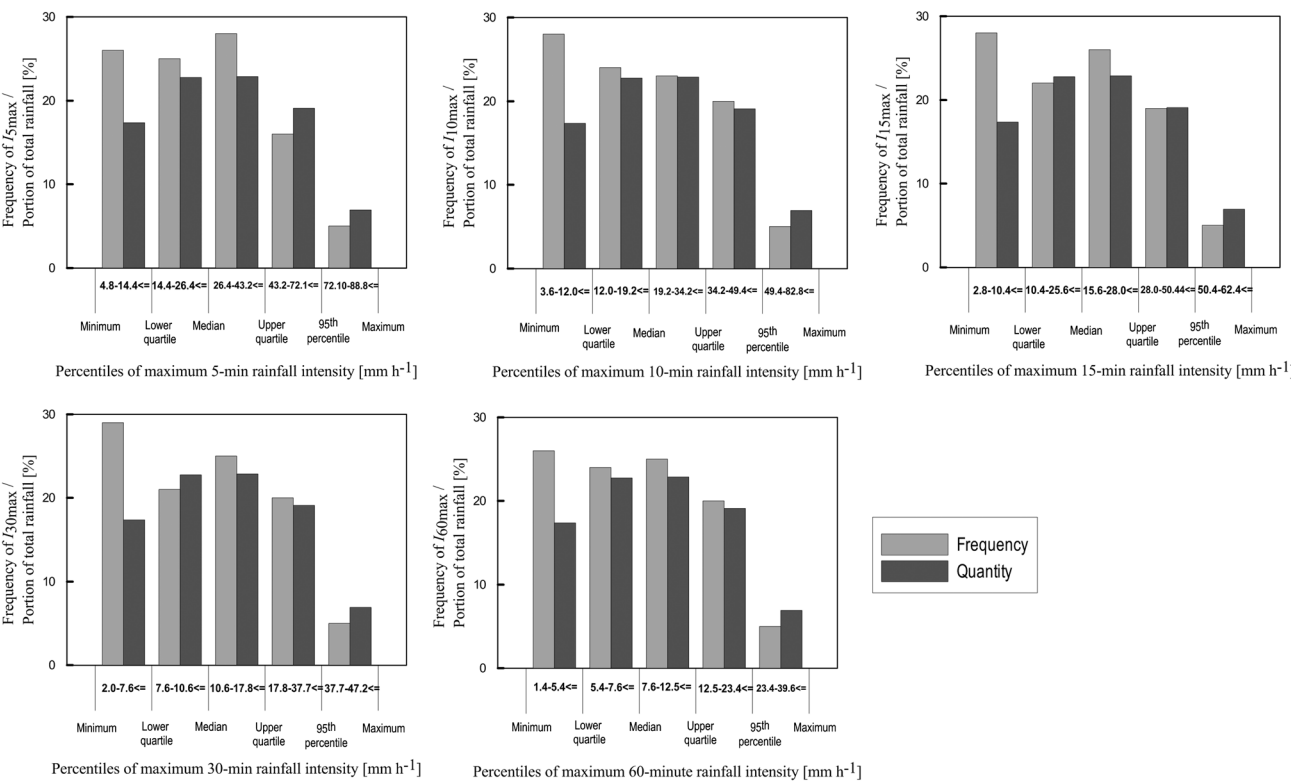


Figure 6. Frequency and quantity of maximum 5-min, 10-min, 15-min, 30-min and 60-min rainfall intensities as recorded at the Dhulikhel rainfall station, Central Nepal, during the 2010 and 2011 monsoon periods.

pasture in Amazonian Perú that had been created by mechanized forest clearing. Pertinently, they are also similar to the 15–20 and 40–45 mm h⁻¹ observed in various Malaysian rain forests underlain by similar soils as the ones found in the Chautara area and subjected to mechanized harvesting using heavy tracked machinery (Kamaruzaman, 1991; Van der Plas and Bruijnzeel, 1993). Such similarities illustrate the advanced stage of

surface soil compaction represented by the Chautara pasture site compared with, for example, the much less intensively used pastures in Amazonia investigated by Zimmermann *et al.* (2006) (median surface K_{fs} ~113 mm h⁻¹) or Tomasella and Hodnett (1996) (66 mm h⁻¹). Such major changes in soil surface K_{fs} after forest conversion and subsequent land degradation are bound to result in important changes in the dominant stormflow

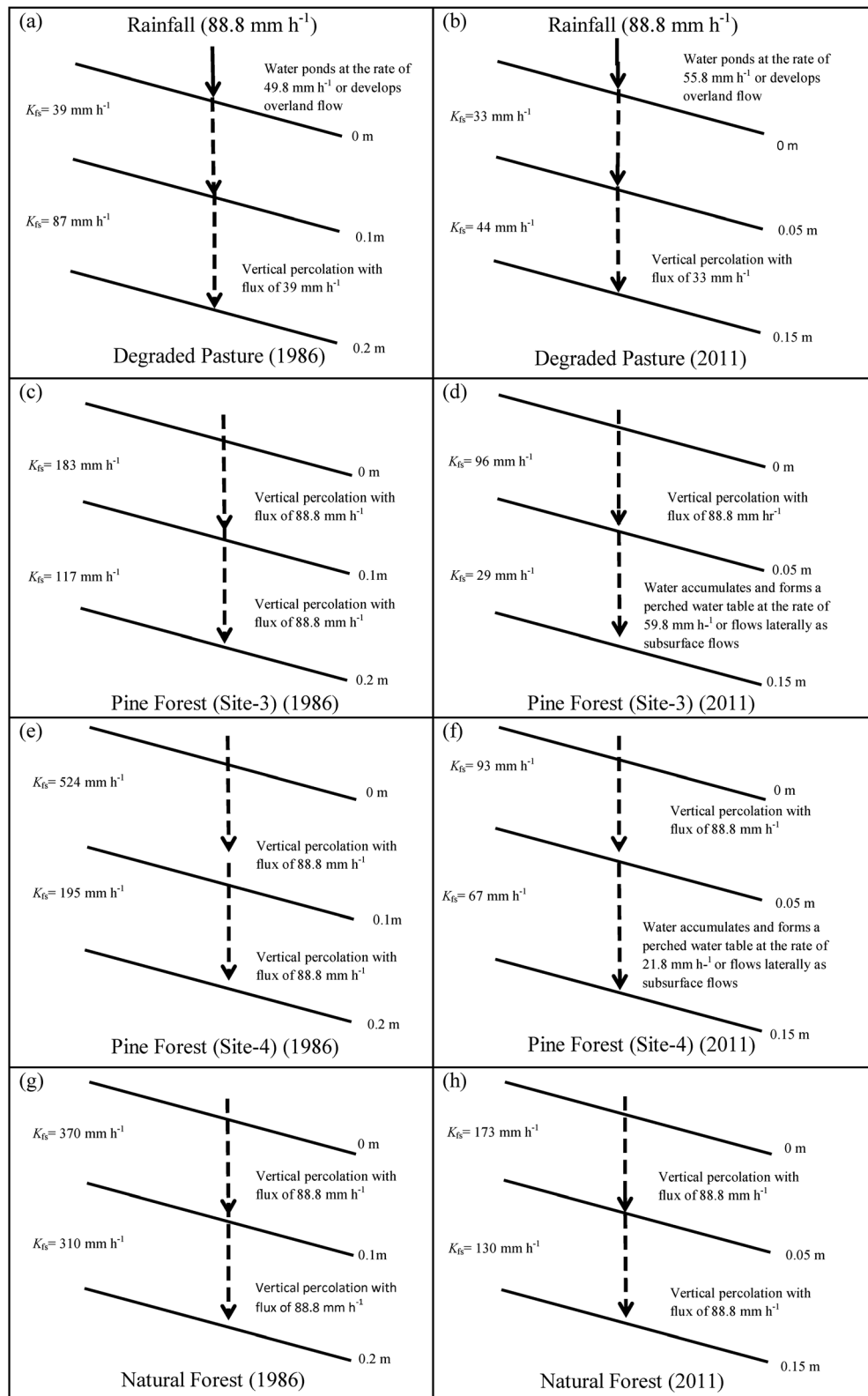


Figure 7. Comparison of the dispersion of rainfall delivered at the maximum 5-min intensity of 88.8 mm h⁻¹ through various soil layers for contrasting land covers near Chautara, Central Nepal in (left) 1986 and (right) 2011. (a,b) Degraded pasture (Site-1), (c,d) Pine reforestation (Site-4) and (g,h) Natural forest (Site-5). K_{fs} is field, saturated hydraulic conductivity. Data for 1986 after Gilmour *et al.* (1987).

pathways over and within hillslopes. Using the data of Gilmour *et al.* (1987) as a baseline, the chief change in K_{fs} at the degraded pasture site was nearly 50% reduction at 0.05–0.15 m depth. The relatively large and statistically significant reduction ($p < 0.05$) at 0.05–0.15 m suggests the

continued trampling and cattle grazing to have a greater influence on the hydrological behaviour of the shallow subsoil than at the soil surface where conditions have possibly reached an extreme stage already, as discussed earlier. IOF is the inferred dominant stormflow pathway as

also suggested previously by Gilmour *et al.* (1987) for this site. Indeed, very high IOF has been reported for similarly degraded bare soils in the nearby Likhu Khola area (57–64% of rainfall; Gerrard and Gardner, 2002), degraded grassland in West Central Nepal (26–47% of rainfall; Impat, 1981) and annually burned grassland in the Philippines (69%; Chandler and Walter, 1998). However, in view of the significant decreases in K_{fs} below the surface (Table III), the generation of lateral SSF at increasingly shallow depths is likely to become more frequent too as soil degradation progresses.

Reforestation. Increases in K_{fs} at shallow depths have been reported by a number of studies in connection with improved soil physical characteristics during natural forest regrowth on former degraded pasture lands in Costa Rica (Deuchars *et al.*, 1999), Amazonia (Godsey and Elsenbeer, 2002; Zimmermann *et al.*, 2006), Ecuador (Zimmermann and Elsenbeer, 2009) and Panama (Hassler *et al.*, 2011). However, comparable studies of changes in K_{fs} in planted forests are still exceedingly scarce (Scott *et al.*, 2005; Ilstedt *et al.*, 2007) and usually comprise a single measurement in time (i.e. at a given plantation age; Mapa, 1995; Zimmermann *et al.*, 2006; Bonell *et al.*, 2010) as opposed to documenting the gradual changes in K_{fs} over time since plantation establishment (cf. Gilmour *et al.*, 1987). As such, the present comparison of K_{fs} values at two points in time separated by 25 years of gradually intensified and changing forest use is a first.

As can be seen from the comparative photographs taken in 1986 and 2011 (Figures 2 and 3), the surface conditions in the two reforestation stands (Site-3 and Site-4) have undergone dramatic changes over the last 25 years, and this is most likely the main reason for the observed radical reductions in surface and near-surface K_{fs} (Table III) and the associated change in inferred dominant hillslope stormflow pathways. Although values of surface K_{fs} in the two pine stands are still sufficiently high to accommodate maximum observed rainfall intensities (Figure 5), the significant reductions ($p < 0.05$) in K_{fs} observed at shallow depth (0.05–0.15 m) at Site-3 in particular (down to 29 mm h^{-1} ; Table III) suggests that the occurrence of near-surface SSF will become much more frequent, even to the extent that widespread SOF may occur during prolonged events of sufficiently high intensity (cf. Bonell *et al.*, 1983). SOF is less likely to be generated at Site-4, where K_{fs} at 0.05 m depth is still sufficiently high at 67 mm h^{-1} to accommodate all but the most intensive 5-min rains (Table III and Figure 5). However, the fact that K_{fs} was reduced about threefold within less than three decades does not bode well for the future unless protective measures can be taken (see succeeding text). The relatively larger decadal reduction in K_{fs} at 0.05–0.15 m depth at Site-3 compared with that derived for Site-4 (Table III) indicates the much more substantial and widespread compaction of the surface soil by human trampling and cattle grazing at Site-3 relative to Site-4, as also evidenced by the much greater bulk density of the topsoil at Site-3 (Table I). Furthermore, in contrast to Site-3, the forest of

Site-4 still had a reasonably developed understory, although grazing was largely absent (cf. Figures 2 and 3). Forest disturbances such as the removal of understory vegetation for fodder or fuel wood, the regular collection of litter from the forest floor and grazing by cattle all have an adverse effect on soil faunal activity and biomass (Lal, 1988; Ding *et al.*, 1992) and therefore on the incorporation of organic matter (cf. Table I), with adverse knock-on effects on soil aggregate stability, macro-porosity and pore connectivity (Lal, 1987; Hairiah *et al.*, 2006). The removal of leaf litter is particularly harmful in which it also leads to greater exposure of the bared soil to the erosive force of drip from the canopy that is typically much greater than that of rainfall in the open, particularly in broadleaved forests (Wiersum, 1985; Brandt, 1988; Hall and Calder, 1993). Further, infiltration capacities of soils from which litter has been removed repeatedly are generally much reduced, whereas IOF production tends to increase dramatically (Tsukamoto, 1975; Wiersum, 1985; cf. Gerrard and Gardner, 2002; Zhou *et al.*, 2002; Hairiah *et al.*, 2006).

The present findings highlight the need for some form of protection of reforested areas to enable the forest soils to realize the enhanced infiltration/percolation benefits envisaged at the time of reforestation. Otherwise, any benefits from replanted forests and the added incorporation of organic matter into the soil (cf. Gilmour *et al.*, 1987) are negated by the sustained biomass harvesting and surface deterioration, as shown by the dramatic decreases in near-surface K_{fs} over the last 25 years inferred by the present study. The human dimension is commonly less emphasized in the tropical forest hydrological literature (Tomich *et al.*, 2004a; Bonell and Bruijnzeel, 2005) or in studies of the hydrological impacts of forestation (Scott *et al.*, 2005; Ilstedt *et al.*, 2007). However, in view of the intensive use of reforested lands in the study area (Figures 2 and 3) as well as in other densely populated uplands in Asia (e.g. Singh *et al.*, 1984; Wiersum, 1985; Ding *et al.*, 1992), it is clear that consideration needs to be given to striking a balance between utilization of tree and forest products and overall landscape rehabilitation. This is a challenge for local communities who need to manage their common land to optimize the provision of goods and services, and it requires a sophistication and maturity of local management that is not yet present in Nepal. Criticism is often voiced about the use of pines in reforestation programmes, as most farmers prefer products from broadleaf species rather than pines for inputs into their farming systems (Gilmour and Fisher, 1991; cf. Singh and Singh, 1992). However, the ecological reality is that most broadleaf species do not survive when planted on the degraded sites that are generally available for reforestation in the Nepalese Middle Mountains (Applegate and Gilmour, 1987; Gilmour *et al.*, 1990). Pines, as pioneer species, can survive and grow in such harsh conditions, which can then provide the site amelioration needed to allow broadleaf species to grow as an understory (Singh and Singh, 1992) that can subsequently be managed to dominate the forest if the local CFUGs decide to manage their forests in this direction (Gilmour *et al.*, 1990; Gilmour

and Fisher, 1991). Conclusions of watershed-wide impacts of reforestation need to be drawn in the light of watershed-wide changes in land use over time; and in the case of the Nepalese Middle Mountains, these changes have been in several directions. As mentioned in earlier sections, there has been a substantial increase in tree and forest cover on both private and common land across most of the Middle Mountain Zone during the past several decades as well as a substantial decrease in open-range grazing. A more recent phenomenon is the large scale and on-going abandonment of agricultural land – both irrigated and rain-fed terraces – as a direct result of the high levels of out-migration, particularly of the active male labour force, to urban areas and overseas (Kollmair and Hoermann, 2011; Gilmour and Shah, 2012). The extent of this agricultural abandonment is not yet clear, but indications are that as much as one-third of all agricultural land may have been abandoned in many districts (Paudel, *et al.*, 2012). Some of this abandoned land is regenerating naturally to trees, further adding to the increasing tree cover across the region. The overall impact of these changes on the hydrological functioning of catchments in the Middle Mountains is as yet far from clear (see also Implications of diminishing soil hydraulic conductivity for dry-season flows section). What is clear is that substantial changes have taken place in recent decades and changes are still underway. The situation is extremely dynamic and multi-faceted.

Gradually degrading natural forest. Even though the transformation between 1986 and 2011 of a nearly undisturbed natural forest to a shrub land with only scattered remnant trees (Figure 4) was demonstrated to greatly influence surface and subsurface K_{fs} (Table III), vertical percolation still remained the dominant rainfall redistribution pathway at Site-5, as it was in 1986 (Gilmour *et al.*, 1987; Figure 7(g–h)). Whilst the main tree canopy had largely disappeared, a dense shrub layer as well as younger trees and an herbaceous layer remained in place (Figure 4). In addition, a thick litter layer was maintained because of the exclusion of cattle grazing and litter collection. The present results illustrate the overriding importance of a well-developed litter layer for soil surface protection compared to the role of the understory (Wiersum, 1985). They further illustrate the important point made by Smiet (1987) that the margins for forest management with regard to soil protection are much broader than those associated with non-forest types of land use. The degraded natural forest is still capable of fulfilling its soil protective role despite having lost much of its main canopy, whereas the severe disruption of the litter layer in the pine forest of Site-3 shows that the margins have been exceeded (cf. Smiet, 1987). Nevertheless, the observed trend of significantly decreasing surface (by 53% since 1986) and near-surface (by 58%) values of K_{fs} at Site-5 is alarming. As demonstrated elsewhere in the region (Patnaik and Viridi, 1962; Gerrard and Gardner, 2002; Bonell *et al.*, 2010), the introduction of any extra pressure to this site in the form of cattle grazing and litter collection over the next few decades is likely to greatly reduce the

surface soil's infiltration capacity and affect the dominant stormflow pathway. For instance, in the nearby Likhu Khola area, Gerrard and Gardner (2002) reported IOF to be negligible in largely undisturbed forest, whereas IOF was rampant in degraded and grazed *Shorea* forest (see also various examples from other parts of the Himalaya discussed by Bruijnzeel and Bremmer, 1989). These results again underscore the need for some form of protection of the remaining natural forests of the study area if they are to maintain their original hydrological functions.

Implications of diminishing soil hydraulic conductivity for dry-season flows. The adequate recharge of groundwater reserves during the monsoon period (June–September) is of critical importance for maintaining streamflow during the extended dry season in the Middle Mountains of Central Nepal. Reforestation of degraded pasture lands is often recommended as a means of restoring stable seasonal streamflow because of the enhanced infiltration afforded by improved soil porosity, structure and water retention capacity after tree planting (e.g. Bartarya, 1989; Negi *et al.*, 1998). However, such a notion is conditional on the gain in groundwater recharge through increased rain infiltration being in excess of the increased plant water uptake after reforestation (Bruijnzeel, 2004; Chandler, 2006). Therefore, any reductions in site water losses via overland flow (and thus potential increases in soil moisture replenishment) will play a crucial role when trying to boost diminished dry-season flows. In view of the very high runoff losses incurred via IOF reported for severely degraded grasslands in the region (Impat, 1981; Gerrard and Gardner, 2002), there is much to be gained by improving infiltration by reforestation of such land in the study area.

Even though rainfall intensities were relatively low for the majority of rainfall events during the investigation period (Figure 6), events having relatively high intensities (i.e. ranging from the median value until the absolute maximum) contributed nearly half of the total rainfall received. Furthermore, the presently observed radical and significant reductions in near-surface values of K_{fs} over the last 25 years for the two planted forests suggest that there may be an equally radical shift in the dominant stormflow pathway at the hillslope scale over the next decades if these forests are not better managed. The chief inferred change in hillslope hydrological response to rainfall associated with continued forest degradation is that from SSF-dominated flow to a much more frequent occurrence of IOF. If continued long enough to reach a truly degraded state (as already observed at the pasture site), the extra volumes of water lost as IOF might well begin to critically reduce groundwater recharge during the rainy season (and thus dry-season flows). Furthermore, the gain provided through enhanced infiltration after reforesting degraded pasture land is most likely to be (much) less if the soils are not allowed to recover. In addition, such diminished gains in infiltrated water will be outweighed more rapidly by the increased evaporative demand of the planted forest relative to that of the pasture (cf. Scott *et al.*, 2005). Such a scenario of

gradually diminishing infiltration opportunities through continued exploitation of the forest may, in turn, cause already diminished dry-season flows to be reduced even further (cf. Bruijnzeel, 1989). Thus, continued human usage and cattle grazing in the planted forests will most likely lead to a continued decline in K_{fs} that, in turn, will further contribute towards a reduction in dry-season flows in the future. Madduma Bandara and Kuruppuarachchi (1988) have described such a case in Sri Lanka, where over a period of 35 years, the gradual land degradation associated with small-holder cropping on former well-managed tea estates produced a reduction in dry-season flows of ca. 33% in the Mahaweli River Basin, whereas wet-season flows increased by 45–50%. Assuming an estimated increase in annual water use of (mature) forest plantations over (non-degraded) grassland of 250–300 mm year⁻¹ (based on a mean annual precipitation figure of 1500 mm and the generalized Zhang-curves for forest and grassland water use; Zhang *et al.*, 2001), it follows that IOF under reforestation in the study area would need to be reduced (relative to IOF generated on degraded pasture) by an amount equal to at least 16–20% of the annual rainfall. On the basis of the presently determined values of surface and near-surface K_{fs} for the two pine stands *vis á vis* the degraded pasture, such a situation still prevailed in 2011. However, in view of the observed decline in topsoil properties in the study area and in degraded forests elsewhere in the region (Gerrard and Gardner, 2002), it would appear to be only a matter of time before the balance will swing.

The present findings do point to the need for balancing the socio-economic and hydrological functions of forests – both planted and natural – in densely populated tropical uplands. Potentially promising developments in this regard include a greater emphasis on agro-forests that contain a variety of tree and crop species serving a range of uses as opposed to the mono-specific character of most planted forests (Hairiah and Van Noordwijk, 2005; Gilmour and Shah, 2012) and, although still at an initial stage, various approaches aiming at rewarding the upland poor for environmental services rendered (see the following for details: Van Noordwijk *et al.*, 2001; Tomich *et al.*, 2004b; Murdiyarso, 2005; Swallow *et al.*, 2009; Van Noordwijk and Leimona, 2010).

CONCLUSIONS

The results of the present study concerning the effects of multi-decadal forest use on soil physical characteristics and inferred hillslope hydrological functioning in the Middle Mountains of Central Nepal can be summarized as follows:

1. Abrupt and significant decreases in field, saturated hydraulic conductivity (K_{fs}) were observed within the upper 0.25 m of the soil profiles under both natural and planted forest sites after 25 years of various forms of human activity. This reflects a loss of macro-porosity due to a reduced incorporation of soil organic matter after repeated collections of litter from the forest floor

supplemented by continuous cattle grazing. Contrary to the forest sites, the studied degraded pasture site showed little change in surface K_{fs} , suggesting surface degradation had reached a ceiling. However, K_{fs} of the near-surface layer was affected negatively by the continued trampling and grazing.

2. Despite several decades of human activity, the natural forest still retained a ‘memory’ at all investigated depths from the previous less disturbed state as recorded in 1986. Therefore, vertical percolation is still the dominant hydrological pathway. However, the continued exploitation of the forest, and the introduction of litter collection and animal grazing in particular, may change this unless some form of forest protection is realized in the future.
3. Rainfall events with relatively high intensities (i.e. above the median value) contributed nearly half of the total rainfall received during the two monsoon seasons covered by the investigation. On the basis of these rainfall characteristics, there are indications in the intensively used planted forests of a change from the formerly dominant subsurface hillslope stormflow pathway to more frequent occurrence of infiltration-excess overland flow. If the ongoing process of soil degradation continues over the next few decades, it will most probably result in a further significant reduction in rainfall infiltration and, ultimately impaired groundwater recharge during the rainy season. Such circumstances are likely to impact dry-season flows adversely, also because the extra water use of the planted forest (relative to the degraded pasture that it replaces) is likely to be in excess of the (diminishing) gains through infiltration if forest degradation continues.
4. This study has demonstrated that management of community forests in Nepal’s Middle Mountains needs to include a consideration of hydrological functioning. Simply planting trees on degraded land is not sufficient in itself to restore the hydrological functions of degraded catchment areas. Attention needs to be given to on-going management of the areas to balance product usage with hydrological functioning. Of specific importance are: (i) balancing utilization (particularly the removal of litter, understory material and cattle grazing) with retention of organic matter and its incorporation into the surface soil, and (ii) limiting activities such as animal grazing that can lead to compaction of the soil surface. This represents an added dimension to the already complex process of community forest management and will pose a considerable challenge to both local communities and the Government.

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